西南交通大学学报

第 58 卷 第 4 期
2023 年 8 月

JOURNAL OF SOUTHWEST JIAOTONG UNIVERSITY

Vol. 58 No. 4
Aug. 2023

ISSN: 0258-2724 **DOI：10.35741/issn.0258-2724.58.4.65**

Research article

Education

LEARNING COMPLEX PLANE TRANSFORMATIONS USING A HISTORICAL CONTEXT MEDIATED BY ICT

利用信息通信技术介绍的历史背景学习复杂平面转换

José Rodrigo González Granada [a, b,*], José Gildardo Murcia [b], Fernando Mesa [b]
[a] Universidad Tecnológica de Pereira
Pereira, Colombia, *jorodryy@utp.edu.co*, *femesa@utp.edu.co*
[b] Institución Educativa Manuel Elkin Patarroyo
Pereira, Colombia, *joseqm140@gmail.com*

Received: June 15, 2023 ▪ Reviewed: July 10, 2023
▪ Accepted: August 19, 2023 ▪ Published: August 31, 2023

This article is an open-access article distributed under the terms and conditions of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0)

Abstract

This study focuses on studying and investigating the learning outcomes of tenth-grade students regarding transformations in the complex plane. It does so by incorporating a historical context by using information and communication ICT. To assess the impact of ICT, specifically GeoGebra software, on students' understanding and conceptualization of complex plane transformations, this study employs didactic engineering methodology and Duval's Theory of Semiotic Representations. The hypothesis states that implementing didactic situations mediated by ICT can lead to positive learning outcomes among students. This research contributes to the integration of the complex plane into secondary-level mathematics education while encouraging curricular and methodological innovation that aligns with educational guidelines and standards. By leveraging technology, particularly GeoGebra, the learning process is supported, emphasizing spatial thinking and geometric systems. This study identifies common challenges students face when working with complex numbers and underscores the importance of graphical representations and conversion between different registers. Ultimately, the findings shed light on the potential of ICT in enhancing mathematics education and fostering students' understanding of complex numbers and their transformations.

Keywords: Complex Numbers, Complex Transformation, Information and Communication Technologies, Didactic Engineering Methodology

摘要 这项工作的重点是研究和调查十年级学生关于复杂平面转换的学习结果。它通过利用信息和通信技术纳入历史背景来实现这一目标。为了评估信息和通信技术，特别是地理代数软件对学生对复杂平面变换的理解和概念化的影响，该研究采用了说教工程方法和杜瓦尔的符号学表示理论

。该假设指出，实施由信息和通信技术介导的教学情境可以导致学生的积极学习成果。这项研究有助于将复杂的平面融入中学数学教育，同时鼓励符合教育指导方针和标准的课程和方法创新。通过利用技术，特别是地理代数，支持学习过程，强调空间思维和几何系统。该研究确定了学生在处理复数时面临的常见挑战，并强调了图形表示和在不同寄存器之间转换的重要性。最终，研究结果揭示了信息通信技术在加强数学教育和培养学生对复数及其转换的理解方面的潜力。

关键词: 复数、复杂变换、信息和通信技术、说教工程方法

I. INTRODUCTION

Education in the modern era faces challenges posed by technological advancements and societal demands, requiring active citizen participation in its construction. Within the field of mathematics education, there is a need for pedagogical strategies that improve the quality and accessibility of knowledge by leveraging information and communication technologies (ICT).

In Colombia, the Ministry of National Education promotes the enhancement of teaching practices through its program of scholarships for teaching excellence. As part of this program, a classroom-based research proposal is presented, focusing on mathematics education, specifically transformations in the complex plane, with tenth-grade students from an educational institution in Dosquebradas, Risaralda, Colombia.

The main objective of this study is to develop a didactic proposal on transformations in the complex plane, using educational software and historical context as motivational tools for learning. The aim of this study is to establish a connection between algebra, in terms of complex number transformations, and geometry, enabling the visualization of algebraic concepts on the plane. Special emphasis is placed on transformations that preserve the distance between two points, which are known as isometries in the plane. This theme falls within the domain of spatial thinking, one of the various types of mathematical thinking.

The primary goal of this study is to create a didactic proposal that promotes spatial thinking and improves students' performance in mathematics through the study of transformations in the complex plane.

II. PROBLEM STATEMENT

Based on the guidelines provided by the United Nations [1], the 2030 Agenda, and the Sustainable Development Goals (SDGs), specifically Goal 4 to ensure inclusive, equitable, and quality education and promote lifelong learning opportunities for all [1], it is intended that education equips students with the necessary tools to navigate different contexts with knowledge relevant to their lives. The focus on quality lies in ensuring that knowledge acquisition enables the assertive appropriation of knowledge to participate effectively within one's vital context.

In recent years, research in mathematics education has been evolving, resulting in various research methods that address the issues faced by educational institutions. In line with this, the topic of transformations in the complex plane has been chosen as a means of understanding the concept of movements in the plane. Historical situations will be incorporated to motivate students and stimulate their learning. The aim is to develop a well-rounded competence in this area.

This study will approach the research using didactic engineering methodology, which is known for its methodological characteristics rather than the specific research objectives it encompasses [2]. This approach offers various possibilities for teaching mathematics at the school level.

Indeed, mathematics education requires innovative strategies that align with the constant updating of knowledge, allowing for a didactic approach. Drawing on a cognitive approach and focusing on the characteristics of cognitive processes associated with the use of representations, as proposed by Duval's Theory of Semiotic Representations, Aznar et al. [3] provide an analysis of the conversion between graphical and algebraic representations of complex numbers among university students. They conclude that systematically addressing this conversion task is essential to promote coordination between different representations and conceptualization of the mathematical object under study.

Numerous studies confirm that the use of technology aids students in interpreting mathematical concepts by providing increased access to multiple representations, which facilitate the articulation between different representations and allow students to work

individually, verify their ideas, and test their problem-solving results [4].

Given the aforementioned context, the following research question is posed: What are the aspects resulting from the learning achieved by tenth-grade students in transformations in the complex plane, using a historical context mediated by ICT, within the didactic situations model?

Therefore, the hypothesis focuses on the idea that the implementation of didactic situations mediated by ICT using GeoGebra software can generate learning outcomes among tenth-grade students at the Manuel Elkin Patarroyo Educational Institution.

III. JUSTIFICATION

This research is relevant to the integration of the complex plane in mathematics education at the secondary level, using ICT to promote innovative and constantly evolving learning in response to global changes. It contributes to curricular and methodological innovation by providing a specific structure in the didactic variables for teaching the complex plane, aligned with mathematical education guidelines, standards, and Basic Learning Rights in the Colombian education system. The use of technology is supported to enhance knowledge production at all educational levels. The focus on geometric procedures and the emphasis on spatial thinking and geometric systems align with the mathematical guidelines and standards for tenth-grade students. The project integrates learning elements from Basic Learning Rights, where students explore and describe the properties of geometric places and their transformations using different representations. The use of software will support the learning process, with a specific focus on the complex plane within the geometry context [5].

IV. THEORETICAL FRAMEWORK

In the first study, the objective was to develop the learning of complex plane transformations in tenth-grade students by incorporating historical perspectives and computer tools. The research includes various components such as epistemological analysis, cognitive assessment, textbook analysis, and the design of activity guides. By integrating the historical context, computer tools, and systematic analysis, this study aims to enhance students' comprehension of complex plane transformations.

This study focuses on the development of a didactic sequence for teaching complex numbers in vocational education. The research emphasizes addressing students' difficulties and the lack of attention given to complex variables by some teachers. This study involved an epistemological analysis, textbook analysis and gathering perceptions from mathematics teachers and students. The findings underscore the importance of the proposed approach in addressing students' challenges and highlight the need for greater emphasis on complex variables in high school mathematics education.

We explore the application of GeoGebra software to enhance the learning of geometric transformations through tessellations in secondary education. This research investigates the impact of software on students' learning outcomes and draws on concepts such as technological capacity and autonomous learning. The study concludes that the pedagogical intervention using GeoGebra software was successful and recommends teacher training and the use of digital platforms for effective learning.

These studies contribute to the field of mathematics education by providing theoretical frameworks, methodologies, and software integration for teaching complex numbers, transformations, and geometric concepts. They highlight the importance of addressing students' difficulties, updating teaching practices, and using digital platforms to enhance mathematical understanding and skills. In addition, the use of Mathematica software is explored in the context of activities on regions in the complex plane, emphasizing the mathematical and digital skills involved and the responsibility of teachers to acquire specific software knowledge. The conceptual framework of this research is grounded in the Theory of Didactic Situations, emphasizing the role of creating situations that promote mathematical thinking and problem-solving. The integration of complex plane transformations, didactic engineering, educational software, and digital competencies (Fig. 2) enhances the teaching and learning process and fosters mathematical understanding and digital skills among students.

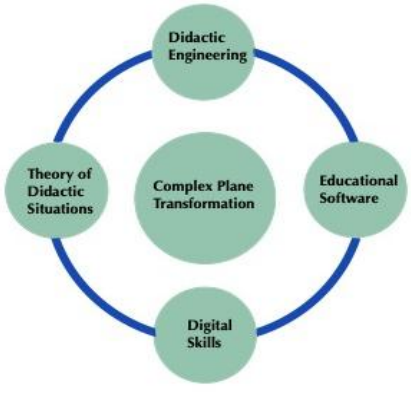


Figure 1. The theoretical foundation of this research

A. Complex Transformation

The concept of complex numbers arises from the limitation of real numbers to contain roots of negative numbers and derivate from quick equations. Complex numbers can express all the roots of polynomials, whereas real numbers cannot. Understanding that complex numbers consist of a combination of real numbers and algebraic expressions is essential. In the complex plane, various movements of complex numbers can be observed, including rigid and inverse movements, translations, rotations, axis rotations, reflections, and slides, as well as transformations in the plane. The historical introduction of the complex plane by Gauss and Argand played a significant role in the acceptance of complex numbers, proving their existence. A transformation in the complex plane is a bijective function that involves considering the images of special sets such as lines, circles, triangles, and squares and observing the modifications they undergo. Rotation on a unit circle and center at the origin, rotation and contraction within a circle with a radius less than 1, and rotation and dilation outside a circle with a radius greater than 1 are common types of transformations. In the complex plane, there is a distinction between the real and imaginary parts of a number, represented as $Re(z)$ and $Im(z)$, respectively. The complex conjugate of z is denoted as $\bar{z}$ or, where the imaginary part excludes the imaginary unit i. The product of a complex number and its conjugate is always real and non-negative. This knowledge facilitates the calculation of the inverse of a complex number. The representation of a complex number is given in Cartesian coordinates for the real and imaginary axes.

B. Didactic Engineering

Didactic engineering is a research technique that focuses on classroom instruction and involves the development, implementation, observation, and analysis of instructional sequences. It emphasizes internal validation through the examination of a priori and a posterior analyses, rather than external validation through experimental control groups. The perception of the arithmetic as a discipline centered on efficient measurement and symbol manipulation hinders deep understanding. Didactic engineering provides a pedagogical framework for comprehensive and concrete learning, involving the formative strategies proposed by the teacher and cooperatively managed by students. It requires a complex system that incorporates action into the curriculum and uses innovative possibilities such as ICT. In mathematics education, didactic engineering serves as both a teaching material production method and a research methodology. It involves interventions in real classroom contexts, using exploratory and interpretative approaches. The roles of the teacher and researcher overlap, and observation and analysis play a crucial role in contrasting the implemented curriculum with students' learning outcomes. The emphasis is on providing students with an authentic learning experience within the classroom setting.

C. Situational Didactics

The Theory of Situational Didactics [6], inspired by Piaget's constructivist perspective, focuses on the artificial genesis of mathematical knowledge through the design of classroom sequences. It recognizes that learning occurs through the learner's adaptation to challenging situations, which stimulates the development of new responses. By creating an environment that generates authentic mathematical problems, teachers facilitate the learner's independent exploration and production of mathematical knowledge. Didactic situations characterized by interactions between students and problem situations, serve as the foundation for formative interactions and knowledge acquisition. These interactions unfold within the processes of abstraction, enabling students to appropriate mathematical concepts and methods.

D. Digital Competencies

To develop digital competencies, it is important to understand the digital characteristics of the NET Generation, which includes individuals who are technophiles and consider themselves "owners" of ICT. They possess remarkable adaptability in activities involving ICT, exhibit a consumerist appetite for new technologies, and are open to change, new behaviors, and social relationships. They are predominantly active, with an integrative, visual, auditory, and kinesthetic nature of learning through non-traditional pathways using new technologies. They have a higher level of visual or iconographic coding and can simultaneously attend to multiple tasks [7].

This characterization aligns favorably with the profile of tenth-grade students in the educational institution where the pedagogical intervention will take place, as it supports the technological framework that will be exploited through the Geogebra software. The project will focus on the technological landscape in addressing complex

topics, understanding that ICT serves as a facilitator and transmitter of information and educational resources for students. Specifically, the project will refer to the use of technologies for learning and knowledge as facilitative instruments for learning and knowledge dissemination [8].

In terms of digital competencies, the project will prioritize informational and technological competencies, emphasizing knowledge, skills, and abilities to work with information, use and handle technological tools, develop multiple literacy as critical viewers and creators of digital publications, engage in cognitive processes to distinguish useful information for knowledge creation, and exhibit digital citizenship competencies to be critical, free, integrated citizens capable of functioning with norms and standards of conduct in both real and virtual spaces [9].

Furthermore, the project strengthens students' logical-mathematical intelligence, which involves effectively using numbers and reasoning. This includes sensitivity to patterns and logical relationships, assertions, and propositions, functions, and other related abstractions. The processes employed encompass categorization, classification, deduction, generalization, calculation, and hypothesis testing [10]. By incorporating these reasoning processes, the project intends to exponentially enhance and develop students' competencies, yielding favorable academic outcomes and performance.

E. Educational Software

The learning tool to be used is educational software, which is currently considered a program of instructions through which users have the advantage of experiencing self-learning on a particular topic by navigating its content. It is also referred to as a curriculum support program because it reinforces, supplement, or serve as pedagogical material in one or more subjects. Educational software can be characterized not only as a teaching and learning resource but also as a specific teaching strategy. The use of particular software entails implicit or explicit application strategies such as exercises and practice, simulation, tutorials, individual or group use, and competition [11].

By using technological tools, the curricular structure of mathematics teaching can be reinforced, particularly in the context of complex planes and the representations of their functions. GeoGebra, freely accessible mathematical software, dynamically integrates geometry, algebra, statistics, and calculus into graphic, analytical, and spreadsheet registers. It offers a variety of tools and specialized calculators that can be used online in real time or downloaded for installation on personal computers. GeoGebra also provides a user-friendly learning community where users can create accounts to access various resources, materials, and downloads [12].

V. Research Methodology

This research is based on the Mixed Method (MM), which combines qualitative and quantitative approaches. This allows for the integration of both methods and facilitates the characterization of integration and underlying logic, leading to a more valid analysis of mixed studies [13]. The qualitative segment will focus on tenth-grade students' perceptions, while mathematical representations will be quantitatively analyzed. The MM includes triangulation, combination, and complementation of methods, collaborating, and confronting results to achieve logical determination and independence [13]. Mixed designs contribute to understanding and transforming educational aspects, enriching the field through the depth and pedagogical transformation of practices [14]. Qualitative data become relevant to capture participants' perspectives, complementing numerical data [14]. This research adheres to didactic engineering, which involves controlling situations through analysis and specifying variables and behaviors [15]. The research design includes four phases: preliminary analysis, conception and a priori analysis, experimentation, and a posterior analysis and evaluation, offering comprehensive learning opportunities [15]. The preliminary analysis considers students' prior knowledge, traditional teaching, their conceptions, difficulties, and constraints [2]. The conception and a priori analysis focus on macrodidactic and microdidactic variables for organizing engineering [2]. The experimentation phase involves implementing engineering with students, with explicit objectives and conditions [16].

- Establishment of the didactic contract.
- Application of research instruments.
- Recording of observations made during experimentation [16].

During this phase, pedagogical processes are executed in line with the set training objectives, supporting the topics covered and referenced in the mathematics curriculum.

Phase 4, a priori analysis and validation, is based on the "set of data collected throughout the experimentation, including observations made during teaching sequences, as well as students'

in-class or out-of-class productions" [2]. This phase contrasts and assesses the pedagogical production carried out through the evaluation and identification of improvement opportunities.

Second, the techniques and instruments used for data collection are addressed. The quantitative data collection technique used was a survey, which can be written or oral. In this study, a written questionnaire was used. Questionnaires were administered to both teachers and students. Students were given pre- and post-tests to investigate their prior knowledge of complex numbers, complex planes, transformations or motions in the plane, figures in the plane, plane isometries, and their manipulation of GeoGebra software and ICT.

Regarding the teacher, questionnaire focused on the didactic methodologies applied for teaching and learning processes related to complex numbers and transformations in the complex plane. The same pre-test administered to students was also administered to teachers. In addition, qualitative methodology employed individual interviews, direct observation of situations occurring during classes, and students' positioning toward the studied mathematical objects, particularly transformations in the complex plane, and their approach to the use of GeoGebra software or ICT as learning tools for mathematical concepts.

A. Phases

1) Phase I. Preliminary Analysis

This Phase I of didactic engineering, according to Artigue [2], comprises three dimensions to address in order to complete the preliminary analysis of the educational fact under study: the epistemological, cognitive, and didactic dimensions.

a) Epistemological Analysis of Transformations in the Complex Plane

The origin of mathematics is historically related to solving problems, some of which were linked to everyday situations that inspired brilliant mathematicians, leading to the development of theories that have brought great advances to humanity. However, there have always been mathematical problems and concepts that are abstract and difficult to comprehend, posing a "headache" for students and a challenge for teachers. The topic of complex numbers is one such example.

Rivero Mendoza [17] noted that imaginary or complex numbers first appeared in the book "Ars Magna" by Giroldamo Cardano in 1545 in Italy when he proposed a solution to the following equation:

$$x^3 = 3p\,x + 2\,q \quad (1)$$

This solution, given by the Scipione del Ferro-Tartaglia-Cardano Formula [17], is named after its three creators.

Cardano's ideas about complex numbers were not accepted by mathematicians of the time and were forgotten for three more decades until Rafael Bombelli, attracted by Cardano's work, continued studying them and introduced the concept of the number -1. However, they were still not accepted by the mathematicians of the time, who considered them unreal, not belonging to this world, and labeled them as imaginary and impossible [17].

In the late 17th-century, the English mathematician J. Wallis made the first approach to the geometric interpretation of complex numbers, which was later concretized independently by C. Wessel and J. Argand in the early 19th-century. This resulted in the geometric representation of the complex number $z = a + bi$ in the Cartesian plane, which gave complex numbers meaning and prevented them from being seen as something impossible. The renowned mathematician Carl F. Gauss played a crucial role in establishing their place in mathematics through his research, in which he clearly presented the properties of numbers in the form $z = a + bi$, known as Gaussian numbers, and their geometric representation. From that point on, the development of complex numbers continued with the contributions of other renowned mathematicians such as Hamilton and Cayley (hypercomplex numbers), Cauchy (differential and integral calculus), and Riemann, emphasizing the importance of complex numbers in the analysis of geometry [17].

Because of the contributions made by these mathematicians and others, complex numbers were defined, along with their operations and corresponding properties, conjugates, modulus and argument, polar and trigonometric form, complex variables, functions, powers, and so on. Their geometric representation and transformations in the complex plane, which are the focus of the present study as the mathematical object to be analyzed in the proposed didactic sequence and its corresponding application in a high school

classroom, were also established.

In other words, in the birth and development of complex numbers, it is possible to identify in the historical epistemological analysis elements of mathematics that allowed mathematicians to transition from one type of thinking to another. These elements include: (a) accepting imaginary quantities as roots of a cubic equation (algebraic thinking of imaginary quantities), (b) using geometric interpretations to make complex numbers more tangible (geometric thinking of complex numbers), and (c) establishing a systematic framework of operations, properties, and representations for complex numbers (symbolic thinking of complex numbers). These shifts in mathematical thinking and the historical development of complex numbers provide valuable insights for the didactic analysis of this topic.

b) *Cognitive Analysis of Transformations in the Complex Plane*

Understanding and working with complex numbers require students to engage in various cognitive processes. A cognitive analysis of transformations in the complex plane involves identifying the mental demands and potential challenges that students may encounter when learning about complex numbers.

One cognitive demand is the need for students to develop a conceptual understanding of complex numbers and their properties. This includes understanding the relationship between real and imaginary components, the geometric interpretation of complex numbers in the complex plane, and the operations and properties of complex numbers. Students need to grasp the idea that complex numbers combine real and imaginary parts and that they can be represented as points in the complex plane.

Another cognitive demand is the ability to perform operations on complex numbers. This involves adding, subtracting, multiplying, and dividing complex numbers, and computing their modulus, argument, and conjugate. Students need to develop procedural fluency to perform these operations accurately and efficiently.

In addition, students need to develop spatial reasoning skills to understand and visualize transformations in the complex plane. They should be able to interpret geometric transformations such as translations, rotations, dilations, and reflections in the context of complex numbers and connect them to the corresponding algebraic operations.

Furthermore, students need to develop problem-solving and critical thinking skills to apply complex numbers in various mathematical contexts. This may include solving equations involving complex numbers, graphing complex functions, and analyzing geometric transformations using complex numbers.

Identifying these cognitive demands helps design instructional strategies that support students' learning and address potential challenges. It is essential to provide opportunities for students to actively engage with complex numbers through hands-on activities, visualizations, and problem-solving tasks that promote conceptual understanding, procedural fluency, spatial reasoning, and problem-solving skills.

c) *Didactic Analysis of Transformations in the Complex Plane*

The didactic analysis of transformations in the complex plane focuses on teaching and learning processes. This includes setting instructional goals, organizing content, selecting appropriate representations and resources, and sequencing activities. The goals should target conceptual understanding and procedural fluency, covering the properties and applications of complex numbers. The content organization should follow a logical progression, building on prior knowledge. Visual and symbolic representations, along with interactive resources, aid understanding. Activities should engage students actively, promote exploration and reflection, and involve various learning approaches and technologies. Students should connect representations, apply concepts, and reflect on their learning.

d) *Pedagogical Approaches and Strategies for Teaching Complex Numbers*

To effectively teach complex numbers, teachers can use various pedagogical approaches and strategies. These include using concrete representations, emphasizing geometric interpretations, using visualizations, engaging students in problem-solving tasks, encouraging multiple representations, emphasizing conceptual and procedural understanding,

integrating technology, promoting collaborative learning, introducing historical and cultural perspectives, and using assessment and feedback. By employing these strategies, teachers can create engaging learning experiences that help students develop a solid understanding of complex numbers and their applications in mathematics.

According to the responses obtained from the surveyed teachers, it is evident that the topic of complex numbers has only been addressed by two (2) out of the six (6) teachers, representing 33.33% of the total. The majority of the teachers (66.67%) do not consider it important for high school students to have a clear understanding of some concepts regarding complex numbers and their applications. Additionally, more than 50% of the teachers did not integrate this mathematical concept into other mathematical topics or practical aspects of life.

e) Cognitive Task Analysis

Cognitive task analysis (ACT) is a qualitative analytical approach that allows for the identification of the cognitive activity underlying an individual or group's achievement when presented with a specific task, as well as the characterization of the task itself. The ACT method consists of a set of methodologies that provide access to cognitive competencies and formal and informal knowledge used in task resolution. It can be applied to any situation, activity, or discipline that requires individuals to follow a series of steps to achieve one or multiple subgoals.

The interpretation of an individual's current capacity in a given scenario is based on prior grounding in the constitutive framework of the task, its perceptual demands, and possible ways to solve the task, including expert performance. This is a distinctive feature of the ACT methodological strategy. The connection between task structure, cognitive demand, possible actions, and actual performance leads to a hypothetical model of an individual's real-time step-by-step mental processing. Each of these research methods represents a degree of ACT based on the use of different theories and units of analysis.

As a result, ACT provides a theoretical and operational model to analyze and explain human cognition based on in-depth research on individuals who adaptively respond to environmental conditions. Newell and Simon [18] suggest a differentiated study of three elements of problem-solving, commonly known as the epistemological levels of ACT: a) task environment, b) question space, and c) analysis of real-time processes. The information analysis tool and the first two factors interact in a complex manner during the problem-solving process.

The specialized environment is defined as the set of objective conditions of the question, the existing connections among them, and the task constraints represented by an "omniscient observer" [19], independent of the problem solver. The problem space is the mental image of the task environment that the problem solver develops in their memory of the structural parts of the problem and their interactions when presented with the task. This representation encompasses the initial and goal states of the problem, the mental operators that can be used to transition from one state to another, and the possible intermediate states of the situation.

To analyze the prior knowledge of the 10th-grade students who are part of the sample in this study and who were exposed to the didactic sequence to evaluate the mathematical component, which focuses on complex plane transformations, a pre-test was administered. The pretest consisted of 11 questions individually answered by each student and lasted for 2 hours.

The pre-test questions evaluated the following mathematical concepts or objects: number sets, placement of numbers on the real number line, ordered pairs and their placement on the plane, solving equations in the complex number set, analysis of perspectives in the circumference (sexagesimal system), axial approximation, movement of plane figures, graphing of plane figures on the Cartesian plane, and homotopy of plane figures. The collected results are presented below in a question-by-question analysis.

Out of the 17 10th-grade students who took the pretest, only 2 had some understanding of the concept of number sets. The majority of students did not have a clear understanding of this concept and were unaware of the existence of the set of

complex numbers, except for one student who mentioned it. The number sets remembered by the students are natural numbers, integers, and rational numbers. They do not identify a number set or associate it with its own elements. Fig. 2 shows some responses to question one of the pre-test provided by the 10th-grade students.

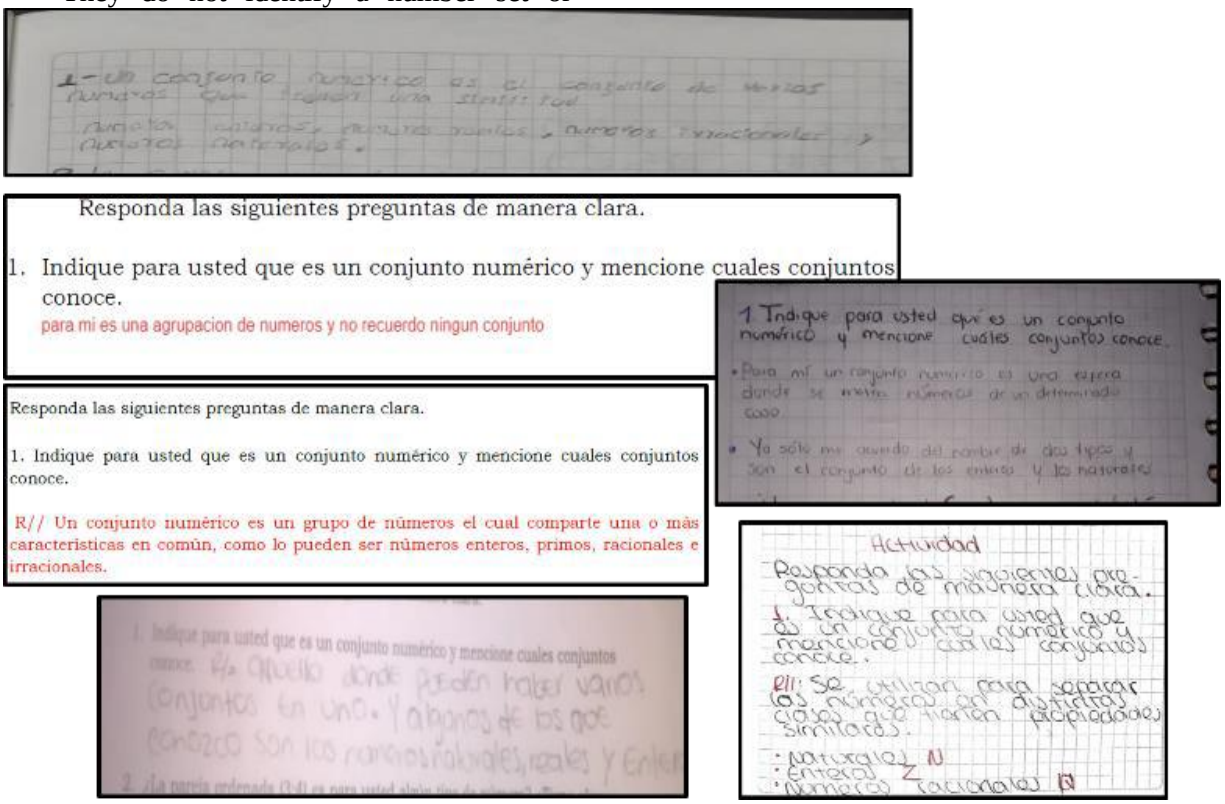


Figure 2. Some responses to question number 1 of the pretest provided by the 10th-grade students

In conclusion, based on the collected results, it can be demonstrated that students comprehend the fundamental geometric characteristics of the proposed exercises. However, when it comes to the analytical aspect, they lack the necessary foundation to provide appropriate responses to the posed questions. Another significant finding is that most students were able to recognize an ordered pair, locate it, and graph it on the Cartesian plane, indicating their understanding of how to locate a Cartesian coordinate and its polar location (implicitly). This prior knowledge will aid in the development of complex numbers and transformations in the complex plane, as outlined in the didactics.

On the other hand, notably less than half of the students recognized the types of movements in the plane and were capable of making comparisons of the same movement among the given figures. However, less than half of them were unable to establish the differences between the presented movements. Furthermore, it is important to consider that students lack prior knowledge of the concept of complex numbers, and to acquire this knowledge, there is a need to expand the range of real numbers.

This allows us to approach the proposed topic through the lens of the Theory of Didactic Situations because, as we can see, mathematical knowledge is not constructed spontaneously but rather requires the design of a didactic experience by the teacher with the aim of creating optimal conditions for students to encounter these new mathematical objects (complex numbers and their transformations in the complex plane) and their own constructions.

2) Phase II. Conception and A Priori Analysis

The a priori examination of a learning environment involves addressing a series of questions aimed at ensuring that the situation has been well constructed and can function effectively [20]. It typically includes an analysis and modeling component that focuses on the qualities of an educational environment intended to be created and the expected outcomes for the students. According to Duoady [21], didactic engineering methodology refers to a series of lesson sequences designed, planned, and articulated coherently over time by a teacher-engineer to perform a specific mathematical learning project for a particular group of students.

Throughout the interactions between the teachers and the students, the program evolves based on the students' responses to the teacher's

decisions and choices. Thus, pedagogical engineering is both a product resulting from a priori analysis and a process that results from the adaptability of applying a product to the variable conditions of a classroom.

This examination is distinct from the experience; as a result, it aims to demonstrate certain disagreements of interpretation that students may manifest, as well as recognize some factors that may affect the acceptable progression of the tasks, considering factors such as: 1) obstacles with the wording of the instructions or the resources used, 2) lack of necessary prior knowledge to tackle the activity, 3) perceptual obstacles, inconsistencies, student questions, and so on.

In this way, the instructor can choose the most effective methodology for the activity to efficiently guide the class. This begins with a priori examination of the set of tasks that comprise the sessions. In summary, the teacher will choose or create a problem related to inference in which they explain and indicate the students' plausible capabilities, allowing them a crucial space to discover, explore, propose, conjecture, justify, and explain their findings throughout the class.

Because of predicting the problems, the teacher will be able to generate questions in advance or assign a new task, so that by helping to resolve a situation, the student will seek incorrect paths, occasionally leading to errors, and the teacher can turn these incorrect decisions into a learning experience. Finally, if a class is planned with the concept of a priori problem analysis in mind, it will be possible to be more competent in class management and build sessions that promote long-term learning through "mathematical meaningful" learning circumstances.

The following 5 sessions were conducted: Introduction to lateral numbers (S0), operations with lateral numbers(S1), use of GeoGebra (S2), transformations in the complex plane(S3), and multiplication of lateral numbers(S4).

These five phases were developed by 37 students, as shown in Table 1ю

Table 1.
Participants in the different sessions

Student Identifier	S0	S1	S2	S3	S4
ST3			*X*		
ST6	*X*		*X*		
ST9	*X*	*X*	*X*	*X*	
ST11			*X*		
ST12	*X*	*X*	*X*	*X*	*X*
ST14	*X*	*X*	*X*	*X*	*X*
ST15	*X*	*X*			
ST16		*X*	*X*	*X*	*X*
ST17	*X*	*X*	*X*	*X*	*X*
ST18	*X*	*X*	*X*	*X*	
ST19	*X*	*X*			
ST21		*X*	*X*		
ST22	*X*	*X*	*X*	*X*	*X*
ST24	*X*	*X*	*X*		
ST27	*X*	*X*	*X*	*X*	
ST28			*X*	*X*	
ST29	*X*	*X*	*X*	*X*	
ST32			*X*		
ST34				*X*	
ST35		*X*	*X*		*X*
ST37	*X*	*X*	*X*	*X*	*X*

3) Phase III. Experimentation

This is the stage of didactic engineering in which the researcher has direct contact with a specific population of students who are the subject of the study. To advance this sequence, the researcher must assert to the students who will be involved in the experimentation the purposes and requirements of the research, as well as maintain a didactic contract and automatically adapt the planned research instruments according to the study topic.

In the same way that the prediction model as a process for teaching mathematics allows for the permeation of traditional algorithmic processes associated with the concept of component because it is the student who proposes the procedures and algebraic expressions that enable them to make assumptions in the didactic circumstance, the use of semiotic registers of representation is given meaning because they are used to demonstrate the sets involved in a relationship.

According to MEN [22], problem-solving is a modeling strategy for situations that give importance and meaning to mathematical objects: "provides the immediate situation in which the mathematical task takes on its full meaning, if the scenarios addressed are connected to everyday experiences and, therefore, are more significant for students" [22]. For the presentation of the experimentation phase, the students were categorized to identify them appropriately. Each participant was considered on the basis of the session they completed. Some participants did not participate or did not complete all the sessions. Therefore, it is assumed that the experimentation phase was tied to the participation of the student population under study.

4) Phase IV. Posterior Analysis and Validation

The experimental phase is followed by the phase of posterior analysis, which is based on examining all the data obtained throughout the experimentation, including observations of

instructional sequences and students' products both inside and outside the classroom. Often, these statistics are complemented with additional information obtained through questionnaires, individual or small group interviews, or other methods used at different moments in the instructional process. Furthermore, as mentioned before, the validation of research hypotheses is based on the confrontation of the two analyses a priori and a posteriori.

In response to skepticism regarding the value of mathematical knowledge acquired through visualization, Guzmán [23] asserts that visualization in mathematics is common, abstractable, and subject to rational and symbolic elaboration, allowing us to deal with the underlying structure of such perceptions more clearly. Geometric education focuses on two main objectives: developing visual thinking and enhancing visualization skills. Due to the importance of visual thinking and visualization, it is necessary to define what is understood by these terms, which are sometimes confused by limited synonyms or related to basic graphics frequently accompanying geometric discourse.

Visual thinking is characterized by verbs such as exploring, selecting, simplifying, extrapolating, analyzing, comparing, concluding, solving, and combining [24]. Visual thinking, as stated by Alsina, Burges, and Fortuny [25], can transform the way geometry is performed and taught if adequately explored. The immediate objective of this work is to analyze the didactic circumstances that, in the words of Brousseau [26], "trigger and produce the problem" in a "game" of potential solutions, some of which are unexpected by the instructor. This demonstrates how a basic difficulty can be a source of learning and can provoke a prior asymmetry in the absence of information.

Mathematical concepts, on the other hand, can be examined from two complementary perspectives: their nature as mechanisms through learning how they function in multiple problems that allow for their resolution, and their nature as cultural objects with a recognized place in the framework of theoretical capacity. This instrument-object dialectic, as emphasized by Douady [21], is crucial in the production of learning situations: a concept often acts as an implicit tool in solving a problem before becoming an established object of knowledge.

Furthermore, concepts can be mobilized in various domains (geometric, numerical, or pictorial), which allows instructors to induce cognitive imbalances in students. In our situation, the utility of the chosen problem goes beyond a restricted understanding; it is a compelling illustration of the various domains mentioned by Douady. Thus, the intention is for the student to engage in tasks that require them to think, explore, make predictions, and verify results, with the aim of faithfully reproducing the mathematical work of a mathematician in the classroom, generating knowledge that can be applied in other situations and transformed into historical awareness.

A good action scenario should enable the student to assess the outcome of their actions and modify them without the interference of the teacher, based on information from the situation's environment. The student interprets the information provided by the circumstance as punishment or reinforcement of their actions. A "conversation" occurs between the student and the circumstance in an action environment. This dialectic of action allows for the enhancement of the implicit paradigm, i.e., having reactions that the student cannot yet describe, verify, or organize into a theory. The student must demonstrate why the concept they have constructed is legitimate in the verification dichotomy.

However, for the student to create a demonstration that makes sense to them, it must be done in a known circumstance called a validation situation, in which they must convince another person. An a-didactic validation circumstance is an opportunity to present a mathematical message (explicit model of the event) as a statement to an interlocutor (opponent). The opponent may request further justifications, rejecting those that they do not understand.

In summary, the posterior analysis phase is a critical step following the experimental phase, in which all the data collected during the experimentation, including instructional sequences and students' work, are examined. This analysis is often supplemented with additional information obtained through various methods such as questionnaires, interviews, and other instructional processes. Validation of research hypotheses relies on the comparison of both a priori and a posteriori analyses.

In response to skepticism regarding the value of mathematical knowledge acquired through visualization, it is argued that visualization in mathematics is a common and abstract process that allows for a clearer understanding of the underlying structure. Geometric education develops visual thinking and emphasizes visualization skills, which should be clearly defined to avoid confusion with basic graphics

that often accompany the geometric discourse.

Visual thinking, characterized by actions such as exploration, selection, simplification, extrapolation, analysis, comparison, conclusion, solution, and combination, plays a crucial role in transforming the way geometry is taught and learned. Understanding the didactic circumstances that trigger and produce problems, even unexpected ones, demonstrates how basic difficulties can serve as sources of learning and lead to a priori asymmetry in the absence of information.

Mathematical concepts can be examined from two perspectives: as mechanisms that allow problem-solving and as cultural objects with recognized positions in theoretical frameworks. This dialectic between instrument and object is instrumental in creating effective learning situations, as concepts often act as implicit tools before becoming established objects of knowledge.

Moreover, concepts can be applied across various domains, such as geometry, numerical analysis, or pictorial representation, enabling instructors to induce cognitive imbalances in students. The chosen problem should extend beyond limited comprehension and serve as a compelling illustration of different domains, fostering tasks that encourage critical thinking, exploration, prediction, result verification, and replication of mathematicians' work in the classroom. The goal is to generate knowledge that can be applied in diverse situations and develop historical awareness.

An ideal learning environment allows students to evaluate their actions' outcomes and modify them independently, based on information within the situation. Students interpret the information provided by the circumstances as feedback for their actions, initiating a dynamic "conversation" between the student and the environment. This dialectic of action empowers students to enhance their implicit paradigm, eliciting reactions that they may not yet be able to describe, verify, or organize into a formal theory. To create a meaningful demonstration, students must validate their concepts by engaging in known circumstances in which they must convince others, presenting the mathematical message as a proposition to an opponent who may request additional justifications for understanding (Fig. 3).

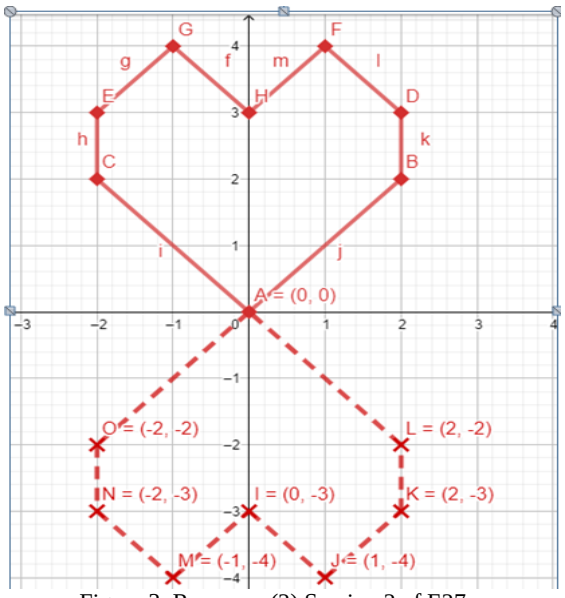


Figure 3. Response (2) Session 2 of E27

In conclusion, the posterior analysis phase and the validation process are essential components of mathematical education research. By examining collected data, including instructional sequences and students' work, and confronting a priori and a posteriori analyses, researchers can gain insights into the effectiveness of instructional methods. In addition, recognizing the significance of visualization in mathematics and promoting visual thinking and visualization skills can contribute to a deeper understanding of mathematical concepts. The chosen problem should extend beyond limited comprehension and facilitate tasks that encourage critical thinking, exploration, prediction, result verification, and replication of mathematicians' works. Creating a supportive learning environment allows students to evaluate their actions, interpret feedback, and engage in meaningful dialogs to validate their concepts (Fig. 3).

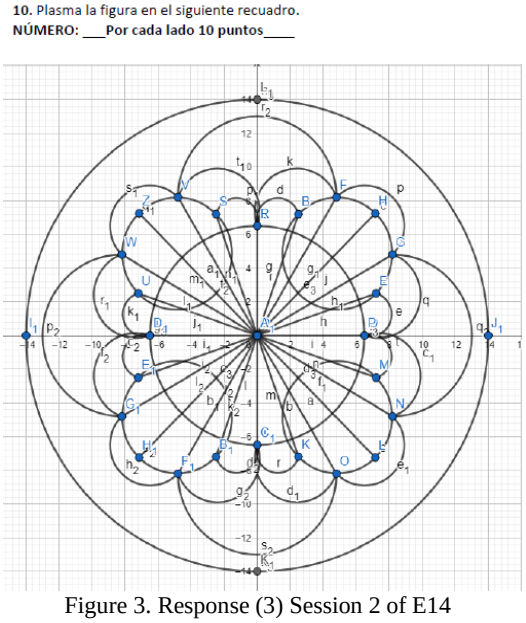


Figure 3. Response (3) Session 2 of E14

VI. Conclusions

To identify the challenges, some doctoral and master's degrees works, as well as scientific articles, were evaluated to understand the documentary sources that described the main shortcomings and difficulties encountered by students when dealing with complex numbers. The following difficulties were highlighted: the failure to connect the history of the object with the use of ICT; a bias toward real numbers over complex numbers in terms of thinking modes; students not having developed a geometric understanding of complex numbers, and the absence of a firm grasp on converting from graphical to algebraic representations in complex number processes.

Additionally, recognized difficulties in the historical and epistemological study were evident when teaching equations with roots that do not belong to real numbers, and discrepancies were found in complex number textbooks, which often presented complex numbers only in algebraic form, neglecting geometric operations. There is no clear concept of teaching strategies, but most teachers agree on the importance of graphical representation. Only one teacher claimed to have used technological resources to teach complex numbers, indicating a lack of integration between complex numbers and other subjects.

Teachers stated a lack of experience in teaching operations with complex numbers in geometric form, while others only explained the Cartesian form, demonstrating limited student work on geometric representations. The use of software as an interactive tool in classroom activities improved student engagement and educational progress, as evidenced by the contributions made and the incorporation of the concepts.

According to Duval [27], "the knowledge of a study relies on the synchronization of at least two registers, and this coordination is manifested by the 'spee'." (The use of software for teaching complex numbers enables a proper transformation between the algebraic and geometric registers, allowing students to understand the meaning of complex numbers).

The Theory of Semiotic Didactics is recognized as a comprehensive theory in the study of complex numbers in their geometric register, as it provides different circumstances at each stage for students to assimilate and develop problem-solving approaches. It promotes individual work and communication, fostering the exchange of ideas and tactics related to the conceptual design phase, providing spaces for organizing information, creating theories, and arguing, corresponding to the validation phase, and ultimately leading to the systematization of learning.

It is crucial to highlight the significance of this work and the uniqueness and relevance of the results obtained.

References

[1] UN (2016) *Sustainable Development Goals*. United Nations. Available from http://www.un.org/sustainabledevelopment/ [Accessed 20/06/23]

[2] ARTIGUE, M. (1995) Ingeniería didáctica. In: ARTIGUE, M., DOUADY, R., MORENO, L., and GÓMEZ, P. (eds) *Ingeniería didáctica en educación matemática*. Bogotá: Grupo Editorial Iberoamérica, pp. 33-59.

[3] AZNAR, M.-A., DISTÉFANO, M.-L., MOLER, E., and PESA, M. (2018) Una secuencia didáctica para favorecer la conversión de representaciones semióticas de curvas y regiones del plano complejo. *Uniciencia*, 32 (1), pp. 46-67.

[4] GONZALEZ, J. (2010) Ideas fuerza que impulsaron la creación de la nueva Universidad Nacional de México y la Escuela de Altos Estudios. *Revista 20/20.*

[5] MINISTERIO DE EDUCACIÓN NACIONAL (2016) *Derechos Básicos de Aprendizaje V.2*. Bogotá: Matemáticas.

[6] BROUSSEAU, G. (1986) Fundamentals and methods of mathematics didactics. *Recherches en Didactique des Mathématiques*, 7, pp. 33-115.

[7] CÓRICA, J.L. and DINERSTEIN, P. (2009) *Diseño curricular y nuevas generaciones: Incorporando a la Generación .NET*. Mendoza: Editorial Virtual Argentina.

[8] CABERO ALMENARA, J. (2015) Reflexiones educativas sobre las tecnologías de la información y la comunicación (TIC). Revista Tecnología, Ciencia y Educación, 1, pp. 19-27.

[9] ADELL, J. (2011) *Cambios en la programación didáctica. Inclusión de la competencia digital. Unidad 3: Nivel básico*. Soledad: Universidad de la Costa CUC.

[10] ARMSTRONG, T. (2017) *Inteligencias múltiples en el aula*. Barcelona: Paidós

Educación.
[11] FERNÁNDEZ I., RIVEROS V., and MONTIEL G. (2017) Software educativo y las funciones matemáticas. Una estrategia de apropiación. *Omnia*, 23 (1), pp. 9-19.
[12] GEOGEBRA (2023) *What is GeoGebra?* Available from https://www.geogebra.org/about [Accessed 20/06/23]
[13] ECHEVARRÍA, J., VOGT, M.E., and SHORT, D. (2017) *Making Content Comprehensible for English Learners: The SIOP Model.* 5th ed. London: Pearson.
[14] PEREIRA PÉREZ, Z. (2010) Los diseños de método mixto en la investigación en educación: Una experiencia concreta. *Revista Electrónica Edcare*, 15 (1), pp. 15-29.
[15] ARTIGUE, M., DOUADY, R., MORENO, L., and GÓMEZ, P. *Ingeniería didáctica en educación matemática*. Bogotá: Grupo Editorial Iberoamérica.
[16] DE FARIA CAMPOS, E. (2006) Ingeniería Didáctica. *Cuadernos de Investigación y Formación en Educación matemática*, 1 (2).
[17] RIVERO MENDOZA, F. (2001) *Una introducción a los números complejos*. Mérid: aUniversidad de los Andes.
[18] NEWELL, A. and SIMON, H.A. (1972) *Human Problem Solving*. Englewood Cliffs: Prentice Hall.
[19] SIMON, H.A., KOZMETSKY, G., GUETZKOW, H., and TYNDALL, G. (1978) *Centralization vs. Decentralization in Organizing the Controller's Department*. Houston: Scholars Book Company.
[20] FURNHAM, A. and CHAMORRO-PREMUZIC, T. (2006) Personality, intelligence and general knowledge. *Learning and Individual Differences*, 16 (1), pp. 79-90.
[21] DOUADY, R. (1995) La ingeniería didáctica y la evolución de su relación con el conocimiento. In: ARTIGUE, M., DOUADY, R., MORENO, L., and GÓMEZ, P. (eds) *Ingeniería Didáctica en Educación Matemática: Un esquema para la investigación y la innovación en enseñanza y el aprendizaje de las matemáticas*. Bogotá: Grupo Editorial Iberoamérica, pp. 61-96.
[22] MINISTERIO DE EDUCACIÓN NACIONAL (1998) *Lineamientos Curriculares para la enseñanza de las matemáticas*. Bogotá: Ministerio de Educación Nacional.
[23] DE GUZMAN, M. (1996) *The Role of Visualization in the Teaching and Learning of Mathematical Analysis*. Madrid: Universidad Complutense de Madrid.
[24] ALSINA, C. and NELSEN, R.B. (2006) *Math Made Visual: Creating Images for Understanding Mathematics*. Washington DC: Mathematical Association of America.
[25] ALSINA, C., BURGES, C., and FORTUNY, J. M. (1988) *Invitación a la didáctica de la geometría*. Madrid: Síntesis.
[26] BROUSSEAU, G. (2007) *Iniciación al estudio de la teoría de las situaciones didácticas*. Buenos Aires: Libros del Zorzal.
[27] DUVAL, R. (1999) Representation, Vision and Visualization: Cognitive Functions in Mathematical Thinking. Basic Issues for Learning. In: *Proceedings of the 21st North American Chapter of the International Group for the Psychology of Mathematics Education, Morelos, 1999*. N.d.

参考文:

[1] 联合国 (2016) 可持续发展目标。联合国。可从 http://www.un.org/sustainabledevelopment/ [访问 20/06/23]
[2] ARTIGUE, M. (1995) 说教工程。在: ARTIGUE, M., DOUADY, R., MORENO, L., 和 GÓMEZ, P.（编辑）数学教育中的说教工程。波哥大：团体编辑伊比利亚-美洲，第 33-59 页.
[3] AZNAR, M.-A., DISTÉFANO, M.-L., MOLER, E., 和 PESA, M. (2018) 一个有利于转换符号学表示曲线和复平面区域的说教序列。合一，32（1），第 46-67 页.
[4] GONZALEZ, J. (2010) 思想力量推动了墨西哥新国立大学和高等教育学院的创建。20/20 杂志.
[5] 国家教育部 (2016) 基本学习权利 V.2。波哥大：数学.
[6] BROUSSEAU, G. (1986) 数学教学法的基础和方法. 数学教学法研究，7，第 33-115 页.
[7] CÓRICA, J.L. 和 DINERSTEIN, P.

(2009) 课程设计与新一代：融入.NET 一代。门多萨：编辑虚拟阿根廷.
[8] CABERO ALMENARA, J. (2015) 关于信息和通信技术的教育反思。科技杂志、科学与教育，1，第 19-27 页.
[9] ADELL, J. (2011) 说教编程的变化。包括数字能力。单元 3：基本水平。孤独: 海岸大学.
[10] ARMSTRONG, T. (2017) 课堂上的多元智能。巴塞罗那：付费教育.
[11] FERNÁNDEZ I., RIVEROS V., 和 MONTIEL G. (2017) 教育软件和数学函数。一种拨款策略。奥尼亚，23（1），第 9-19 页.
[12] 乔格布拉 (2023) 什么是地理代数？可从 https://www.geogebra.org/about [访问 20/06/23]
[13] ECHEVARRÍA, J., VOGT, M.E., 和 SHORT, D. (2017) 使英语学习者的内容易于理解:庇护教学观察协议模型. 第 5 版。伦敦：皮尔逊.
[14] PEREIRA PÉREZ, Z. (2010) 混合方法在教育研究中的设计:一个具体的经验. 电子杂志教育辅导, 15 (1), 第 15-29 页.
[15] ARTIGUE, M., DOUADY, R., MORENO, L., 和 GÓMEZ, P. 数学教育中的教学工程. 波哥大：团体社论伊比利亚-美洲.
[16] DE FARIA CAMPOS, E. (2006) 说教工程。数学教育研究及培训笔记本, 1 (2).
[17] RIVERO MENDOZA, F. (2001) 复数的介绍。梅里德：安第斯山脉大学.
[18] NEWELL, A. 和 SIMON, H.A. (1972) 解决人类问题。恩格尔伍悬崖：普伦蒂斯霍尔.
[19] SIMON, H.A., KOZMETSKY, G., GUETZKOW, H., 和 TYNDALL, G. (1978) 在组织财务主任部门时,集中与分散. 休斯顿：学者图书公司.
[20] FURNHAM, A. 和 CHAMORRO-PREMUZIC, T. (2006) 个性，智力和一般知识。学习与个体差异，16（1），第 79-90 页.
[21] DOUADY, R. (1995) 说教工程及其与知识关系的演变. 在：ARTIGUE, M., DOUADY, R., MORENO, L., 和 GÓMEZ, P. (编辑) 数学教育中的教学工程：教与学数学的研究与创新纲要。波哥大：团体社论伊比利亚-美洲, 第 61-96 页.
[22] 国家教育部 (1998) 数学教学的课程指导方针。波哥大：国家教育部.
[23] DE GUZMAN, M. (1996) 可视化在数学分析教学中的作用. 马德里: 马德里康普鲁滕斯大学.
[24] ALSINA, C. 和 NELSEN, R.B. (2006) 数学使视觉化:创造图像来理解数学. 华盛顿特区：美国数学协会.
[25] ALSINA, C., BURGES, C., 和 FORTUNY, J. M. (1988) 邀请几何学的教学法。马德里：综合.
[26] BROUSSEAU, G. (2007) 说教情境理论研究导论。布宜诺斯艾利斯：画眉的书.
[27] DUVAL, R. (1999) 表示，视觉和可视化：数学思维中的认知功能。学习的基本问题。在：国际数学教育心理学小组第 21 北美分会会议记录，莫雷洛斯，1999 年。没有数据.